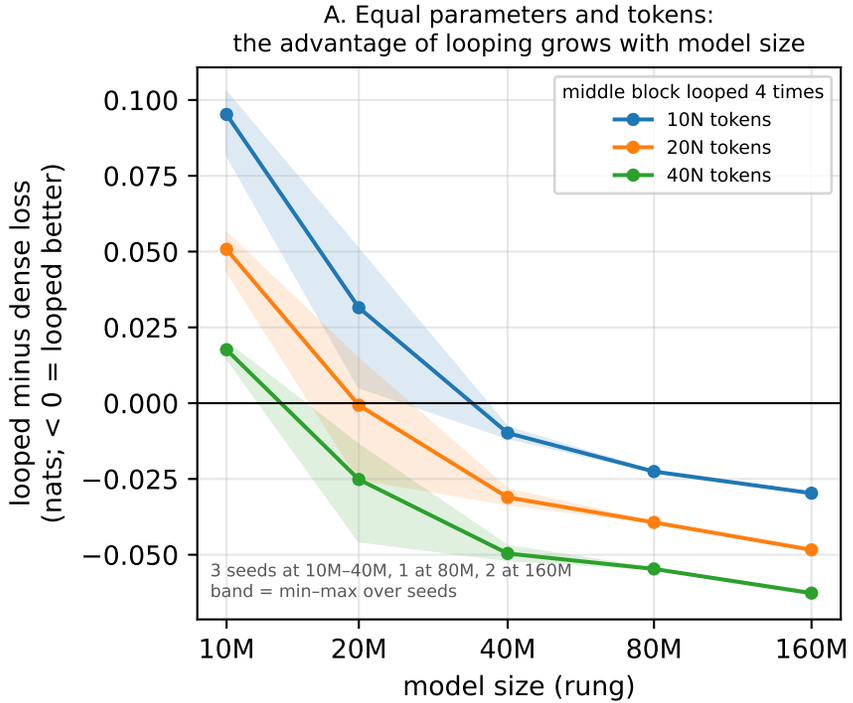
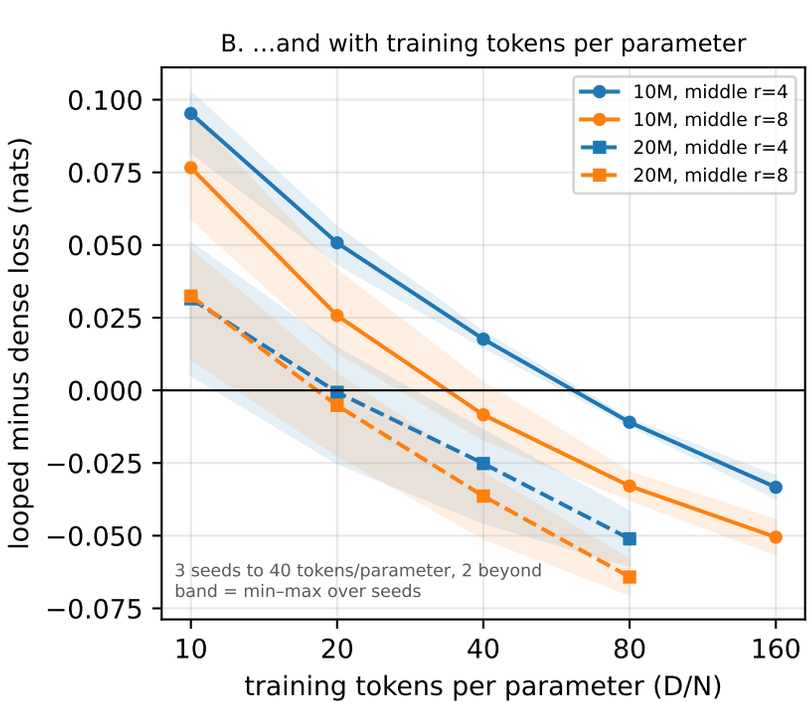
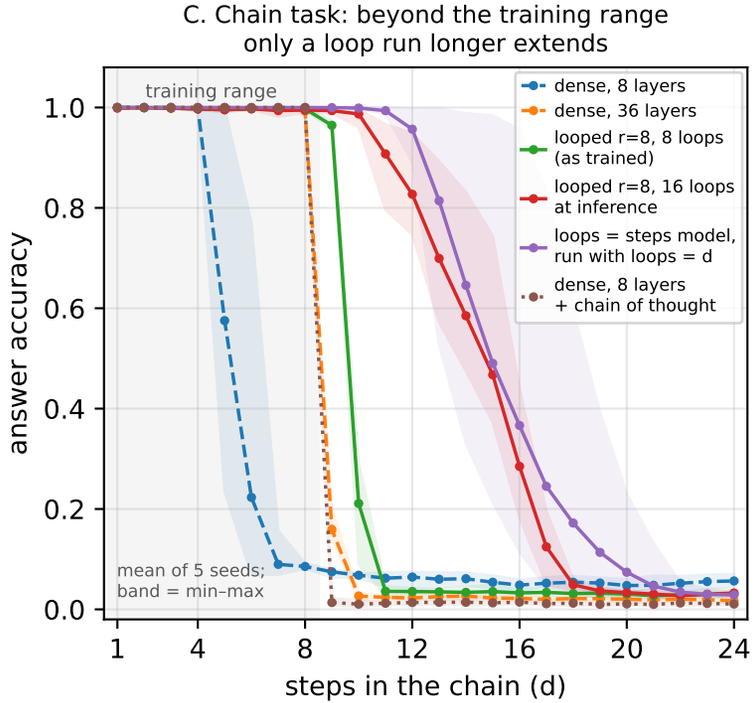

A. Equal parameters and tokens:
the advantage of looping grows with model size
looped minus dense loss
(nats; < 0 = looped better)
middle block looped 4 times
10N tokens
20N tokens
40N tokens
3 seeds at 10M–40M, 1 at 80M, 2 at 160M
band = min–max over seeds
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
10M
20M
40M
80M
160M
model size (rung)


B. ...and with training tokens per parameter
looped minus dense loss (nats)
10M, middle r=4
10M, middle r=8
20M, middle r=4
20M, middle r=8
3 seeds to 40 tokens/parameter, 2 beyond
band = min–max over seeds
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
10
20
40
80
160
training tokens per parameter (D/N)


C. Chain task: beyond the training range
only a loop run longer extends
answer accuracy
training range
dense, 8 layers
dense, 36 layers
looped r=8, 8 loops
(as trained)
looped r=8, 16 loops
at inference
loops = steps model,
run with loops = d
dense, 8 layers
+ chain of thought
mean of 5 seeds;
band = min–max
1.0
0.8
0.6
0.4
0.2
0.0
1
4
8
12
16
20
24
steps in the chain (d)